

HIGH FRONTIER 4 ALL MODULE 4:

EXODUS

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High Frontier 4: Module 4 - Exodus, for 1 to 6 players. Updated Sept 6, 2021.

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4A. Module 4

As Factions settle the solar system and adopt a commodity economy based upon isotope rocket fuel, humanity changes and augments itself to adapt to the space environment of low gravity and high radiation.

The science of cybernetics allows human-machine hybrids to flourish. Terawatt technology makes it possible to build a starship and augmented humanity makes it possible to expand to the stars. **Module 4 Exodus** plays with multiplayer and solitaire scenarios and only needs the Core **High Frontier 4** game to play, but can be combined with any Module. It expands **High Frontier** in scope but not in length:

- a. **Contracts (All Games).** Enhance the early to mid-game, granting early Aqua and mission goals.
- b. **New Spaceborn Colonist Queue (Modules 1 & 2)** represent Humans born in space whose bodies have atrophied to the point that they are largely mechanical.
- c. **Augmentation Chits (Modules 1 & 2)** are gained by completing contracts and played on Colonists to upgrade them and allow them to Promote and be Promoted.
- d. **Enterprises (Modules 1 & 2)** are another path to augmentations, as humanity alters itself and its machines to better colonize the solar system and reach the stars.
- e. **New Solitaire Political Diagram (Module 0)** to integrate contracts with the CEO and altruism solitaire variants.
- f. **An Isobank**, founded by one player and holding the game's repository of isotope fusion fuel, becomes the center of starship construction. With Module 3, the isobank can be raided and potentially robbed! If a starshot occurs, either in the spirit of expansion¹ or to escape from bondage,² a combination game into *Interstellar* is possible.

¹ WHY GO AD ASTRA? Each of us has a drive to spread one's wings that has been reinforced by eons of natural selection. This instinct, rather than "population pressures", will impel a pioneering minority to the stars. Earth is never going to run out of living space, as this is being written, the global population increase is plummeting at a rate such that the population max will be achieved this century. Meanwhile, technological advances continue to shrink the ecological footprint, the number of hectares per person that each of us requires for our food and other needs. A simple application of arcology and high tech farming would provide more living space than the vastness of space could provide for centuries, because of what it takes to make such places habitable.

² EXODUS refers to a mass departure of refugees, as originally accounted in the second book of the Torah and of the Old Testament. The war-refugee scenarios in this Module describe a postwar interstellar exodus comparable to the flight of the Israelites from slavery in Egypt about 3.5 kya.

EASILY MISSED: Because of the inclusion rule (4E3b), if you have at least one Colonist who has an augmentation chit, you are almost guaranteed a seat in the Starship Exodus. [4A1b]

4A1. Core, Starship, & Exodus Game Lengths

Module 4 Exodus lasts from 4 to 7 solar cycles long. The *Combined Game* leads into the next game *Interstellar*.

a. 4-5 Cycle Core Game (competitive).

- **Modules.** Module 0 must be used, and optionally Modules 1, 2, and 3 (Conflict).
- **Ignore Rules in Blue Font.** Rules in blue font are only for the 6-7 cycle games. Ignore these rules, including *isotope monetization*, *enterprises*, *spaceborns*, *augmentations*, and *Project Exodus*. Include those labeled for *Module 3* (Conflict) if using that Module.
- **Number of Players.** 2 to 6 competitive, or 1 solitaire (**4G**).
- **Game Victory.** See **4E2**.

b. 6-7 Cycle Games (competitive).

- **Modules.** Modules 0, 1, and 2 with Futures, and optionally *Module 3* (Conflict).
- **Number of Players.** 2 to 6 competitive, or 1 solitaire (**4G**). *Solitaire* is not supported if using *Module 3*.
- **Game Victory.** See **4E2**.

c. 5-7 Cycle Scenarios using *Module 3* (competitive).

- **Exodus Conventional War (4F1).** Use *Module 3*, but additionally score per **4E2** and *desperation starshots (4E3b)* are allowed.
- **Exodus Union War (4F2).** Use *Module 3*, but additionally score per **4E2** and *desperation starshots (4E3b)* are allowed. War can break out, with the Spacer Union (including Robots) as the Independence bloc and the others as the Loyalist bloc.

d. *Interstellar Combined Game.* This is the same as a 7-Cycle game, but your accomplishments and augmented Colonists are imported into the next game in the series: *Interstellar*. See **4E4**.

e. *Solitaire.* See **4G**. For victory conditions, including Combined Game solitaire, see **4G4**.

4A2. Module 4 Components

- 1 Drawer Box.**
- 1 Rules Booklet.**
- 1 Starship Stack Placard/Alternative Solitaire Political Diagram**
- 4 Player Aids.**

- e. **36 Contract Cards** which form the 3 *contract* (**4B**) decks.
- f. **6 Timepiece Chits** (1 for each of 6 player colors).
- g. **1 Default Chit**
- h. **1 Isobank/Starship Token.** Place this at the base of a Bernal where the isobank is established.
- i. **6 Exodus Contract Cards**
- j. **10 Spaceborn Colonist Cards**
- k. **6 Robot Colonist Cards**
- l. **6 Child Chits** (1 for each of 6 player colors).
- m. **6 Isotope Monetization / Isobank Stack Cards**
- n. **36 Augmentation Chits** (6 for each shape)
- o. **12 Exodus Medal Chits** (TW Thruster, Promoted Bernal, Promoted Freighter, Commander, 8 passengers)

4A3. Contract Card Anatomy

Contract cards have a blue side and a gold side. The contract decks start blue-bordered side faceup. In games of 6 or more cycles, the deck flips to the gold-side when the *isobank* (**4C2**) is built. Contract cards are auctioned after a *fundraise operation* (**O2b**).

- a. **Name.**
- b. **Mass/Rad-hardness.** Use these when the contract is carried as a Stack Card, see **B2h** and **B2j**.
- c. **Type.**
- d. **When Fulfilling or Fulfilled Abilities.** See **4B2e**.
- e. **Ideology Pair.** Individuality-freedom, honor-unity, or equality-authority.
- f. **Cybernetics.** Represent partner agencies (**4B2k**) and are used for implantations (**4D4c**).³
- g. **Destination.**
- h. **Cash Advance.**
- i. **Late Fee.**
- j. **Victory Points.**
- k. **Tophat/Felony.** See **4B2f**.

³ CYBERNETICS is the science of communication and control systems shared by machines and living things. Such an integration is essential for mastering the high frontier as well as for a starship controlled by both humans and machines.

4A4. Augmentation Chit Anatomy

Augmentation chits have their effects on one side, and the corresponding cybernetics icon for implantation (**4D4c**) on the other. They are placed upon a Colonist card via the *implant operation* (**4D4**).

- a. **Name & Shape.** Triangle (isotope), hexagon (batteries), heart (life-support), jar (hibernation), rectangle (warden), and radiation triad (shields).
- b. **Effects.** See **4D3a,b,c,d,e,f**.
- c. **Cybernetics icon.** This is the required icon for implantation, see **4D4c**.

TIP: Because contracts require boosting, there is more boosting in a Module 4 game and consequently the Player White NASA Faction Privilege (with a boosting bonus) is at an advantage. This Faction is recommended for beginning players. [4A5]

4A5. Module 4 Setup

Setup is per Core (**C**) plus for the Modules used plus the following (skip the blue font for a 4-5 Cycle game):

- a. **Seniority Disks.** Four to seven, depending on the *game length* chosen (**4A1**).
- b. **Contract Decks.** Separate the contract cards with into 3 decks, one for each *ideology pair* (**4A3e**), with the blue-side faceup. Shuffle each deck and place near their ideology pair on the political assembly placard.
- c. **Timepiece & Default Chits.** Place these chits near the Sunspot Cycle placard.
- d. **Starting Aqua.** Contract decks are not patent decks and so they do not affect starting Aqua (**C5**).
- e. **Exodus Contract Deck (Modules 1 & 2).** Shuffle the *Exodus* contracts, randomizing which side of each is faceup. Place them on the starship placard with the *isobank token* (**4A2h**) on top to show the deck is locked.
- f. **New Colonists (Module 2).** Shuffle the new Colonists (spaceborn and Robot), plus all Robots from Module 2, into a new spaceborn queue.
 - **Queues.** *Module 4* uses 2 Colonist queues: earthborn and Robot/spaceborn (**4D1**).⁴ “Earthborn” is slang for a non-Robot Colonist from *Module 2*.
- g. **Augmentation & Child Chits (Module 2).** Each player adds to their Reserves the child chit of their color, and a set of augmentation chits of the 6 shapes.
- h. **Module 3 Conflict.** If playing with this Module, decide if the War will be *conventional* (**4F1**) or *Union* (**4F2**). Then additionally setup per **3A2**. Reminder that this may make the final starshot a desperate one.

⁴ EARTHBORN vs. SPACEBORN. A person born on Earth is acclimated to a 1 gee planet or Bernal. A person who has spent years in a low gravity rocket or world suffers from muscular atrophy to the point where they would need a lot of bionic help to take a vacation on Earth. As the future unfolds, humans may specialize into two kinds of cybernetic augmentations: mental aids, such as an exocortex or wearable computers, and physical aids, such as bionic limbs, locomotion, and life-support, all of which allow acrobatic performance in zero-gee.

4A6. Core/Module 0-3 Rules Changes

- a. **Inspiration (K2a).** This event is applied to both Colonist queues plus the *contract decks (4A5b)*, including the *Exodus contract deck* if it is active (4C2c).
- b. **Contract Fulfillment (add to G).** This new free action moves a completed contract card from a Stack to adjacent to your playmat. *Completed contracts are called enterprises (4B8).*
- c. **Negotiation Summary (add to N).** Possession of contract cards can be negotiated only if unfulfilled. Some fulfilled contract Abilities may be negotiable, if specified. Augmentation chits are not negotiable. Fulfilled contract icons may be shared (see partnership delegates, 4B8).
- d. **Fundraise Operation (O2 of Module 0).** *During this Operation, a partner delegate can be placed on an enterprise per 4B8b in order to increase your portfolio.* As the final step of this Operation, you may cycle a contract deck and may start a *contract auction (4B3).*
- e. **FINAO (H7e).** Failure Is Not An Option cost changes after *isotope monetization (3B5).*
- f. **Isobank Stacks (add to E3).** This is a new Stack created by construction of the isobank (4C). Its location for all players is indicated by the single isobank/starship token.
- g. **Miner Colonist Profession (2C1a).** During a *site refuel operation (I5)*, you may perform one extra refuel per Colocated miner present. The refuels may be of different types (you may receive 1 isotope and 7 water, for instance.)
- h. **Exomigration (2A6).** There are 2 queues instead of 1: earthborn and Robot/spaceborn (4A5e). During any exomigration, you must choose whether to join the *Spacer Union (4D2)* or not. If you do, exomigrate the top card from the Robot/spaceborn queue, but if you do not, take the top card from the earthborn queue. If you ever exomigrate a Robot/spaceborn, you must Decommission all your earthborns and vice versa. Regardless if you hold Robot/spaceborn Colonists, or Human Colonists, your Colonist limit is equal to the number of anchored Bernals (doubled for promoted Bernals, see 2Ca).
- i. **Robot Exomigration (2A6).** If multiple Robots are drawn during a single *exomigration (2C2a)*, choose one to keep and Discard the rest, then shuffle the spaceborn queue.
- j. **Ad Astra Futures (1D1b).** If one of these Futures are accomplished with any TW thrusters, promoted Freighters, and/or promoted Bernals present (whether required or not), set aside the corresponding medals. They are not rewarded during *Exodus scoring (4E3c)*. (The ad astra points replace the medals.) Any isotope present is Decommissioned but set aside per 4C4a so that it can be counted in the endgame.

EASILY MISSED (MODULE 2):

If you fail the Epic Hazard Roll, the Bernal supports are Decommissioned and the Bernal is not Anchored.

[4A6k]

- k. **Promotion (1A5 / 2A3).** Promotion is the same as in Modules 1 and 2, except a Colonist (or Colonists) of yours must be present implanted with a certain number of unique augmentation chits (4D7e).
- l. **Anchoring (2A5).** Anchoring the “Lofstrom Loop” or “GEO Space Elevator” Bernals at a location which activates the “HOME” Ability requires an Epic Hazard Roll (1A6), with the Bernal supports as the unit(s) performing the roll.

4A7. Core/Module 0-3 Rules Clarifications

- a. **Bank (clarification).** This term refers to the Space on the map where you store your Aqua and/or gold beads, *except that gold beads are stored at the isobank if one exists (4C2).* Your Bank is located in LEO, *a Home Bernal (2B3), or (if Module 3) an ET Home Bernal (3F1), whichever is furthest from Earth.* You may only have one Bank at a time.
- b. **Powersat Stacking (clarification).** Modifications to the Powersat push value are limited to +1 maximum (total push +2), except for the MagBeam robonaut (which always gets a total push of +3 thrust regardless of Powersat push value).
- c. **Multiple Rerolls (clarification).** If multiple Abilities and/or rules grant rerolls for a particular roll, you may reroll once per such Ability/rule. After a reroll, you decide whether to keep it or roll again (if possible).
- d. **Site Descriptions (clarification).** Some cards in this Module use new methods of describing Sites:
 - **Unexplored:** A Site without a claim disk or busted disk.
 - **Flyby Space:** A Space with a flyby bonus printed on the map, including *solar oberth flyby (H8e).*
 - **Signposted:** Any Site with a colored route leading to it.
 - **Space Elevator:** A Site connected to an elevator path printed on the map (e.g. phobos), *or to an Anchored GEO Space Elevator (1B9e).*
- e. **Dropstone Sites (Clarification):**⁵
 - **Any Synodic Comet** (multiple locations).
 - **Jovian Himalia Moonlets** (elara, leda, himala, lysithea: 10:10 F).

⁵ TERRAFORMING LUNA. The only world that can be terraformed in High Frontier timescales is Luna. If a catcher station is located on the lunar farside, one unintended consequence of the kinetic energy (possibly augmented by protium fusion) from dropstone impacts would be to give Luna an oxygen atmosphere ten times the atmospheric pressure on Mars. Thus, strangely, a terraformed Luna is caused by pollution rather than deliberate effort. Even if fusion operations were to cease, its atmosphere would linger for centuries.

- **Saturnian Norse Moonlets** (thrymr, ymir, phoebe, skoll, tuttungr: 9:10 G).
 - **Saturnian Moon Methone** (9:00 G).
 - **Uranian Sycorax Group** (setebos, sycorax, prospero: 9:10 H).
 - **Comet.** Any Site with “Comet” in the name or map label.
- f. Ownership (Clarification).** References to tokens and Stack items (cards, FTs, etc.) apply to those that you own or which you have permission to use (through negotiation, Freedom to Roam, Felony, etc.), except:

EXAMPLE [4A7f] The Fullerene Cable Spinner contract allows you to ET Produce C cards at “any Factory”; producing such cards at another player’s Factory is not possible without permission. The Robotic Facility contract requires you to factory refuel at “Your Factory”. Even with permission or during the Freedom to Roam Treaty, you cannot fulfill this contract at another player’s Factory.

- “Any player’s” means any such item regardless of permission.
 - “Your” means only items that are in your Playmat or are in your player color.
- g. Order of Operations (clarification).** When performing multiple steps of adding/subtracting and multiplying/dividing, use the following order:
- **Subtraction**
 - **Division / Multiplication**
 - **Addition**
 - **Round if necessary** (up or down as indicated)

4B. Contracts & Enterprises

Contract decks are made up of contract cards representing available for-hire work from various enterprises. The winning bidder gets instant funding in return for a promise to complete the contract by a specific deadline, with penalties for being late. Fulfill a contract by taking its card to the specified location, and (if specified) returning it. Once fulfilled, a contract grants cybernetics icon(s) and (usually) an ongoing Ability, both representing a continuing partnership with a high-tech enterprise. A fulfilled contract is also worth VP at the endgame.

- a. **Contract Categories** include trillionaire passengers, tourists, megaprojects, military contracts, government grants, science payloads, and robotic research.
- b. **Discard.** This glossary term means to put a card to the bottom of its appropriate deck or queue.

4B1. Contract Decks

Contracts form three or four decks placed around the political assembly and have special rules for bidding, inspiration, and being carried in Stacks.

- a. **Inspiration Event.** Like patent decks, all contract decks are inspired during the *inspiration Event Roll (K2a)*.
- b. **Isobank Contract Upheaval.** Each contract has two sides: blue and gold. As soon as an *isobank (4C2)* exists, flip all three contract decks so the bottom is on top and from now on the more profitable gold side is used.⁶
- c. **Exodus Contracts.** This deck (**4A5e**) is composed of special single-sided contracts which start on the starship placard. This deck starts covered by the *isobank token (4A2h)* to indicate that it is locked (and not affected by inspiration Events) until the *isobank* is established. This removes the token from the deck, so that its top card is now auctionable (**4B3c**).

4B2. Contract Cards

Contract cards have Mass and rad-hardness and are carried in Stacks like other cards in play until their contract is fulfilled and they are stored as enterprises.

- a. **Not Patents.** Contract cards do not count against *academia hand limit (I2a)* and cannot be *sold (I3a)*.

⁶ AQUA VS. ISOTOPES. These two commodity currencies differ in their value on Earth. Aqua (water) is not very valuable on Earth, while isotopes (gold beads) are quite valuable as fuel for fusion powerplants. Aqua has no value on Earth's surface, but does have value in LEO. If you pay the programmers in Aqua, you pay in water orbiting in big cans where the programmers can resell it to outward-bound ships that need propellant.

- b. Side Matters.** When boosting, a contract must be placed on the side (blue-side or gold-side) that was faceup when it was auctioned.
- c. Launch Surety Contracts.** Contract cards are immune to *pad explosion/space debris events (K2c)* or *budget cuts events (K2f)*.
- d. Manned Contracts.** If a contract has the “Humans on Board” icon, it is considered Human (see glossary). This is significant for Glitch Repairs, Felony defense, Glory, [or as a manned target during combat](#). It is a Felony to voluntarily Decommission or Discard a manned contract, and if it suffers voluntary Decommission, you suffer *default (4B6)*.
- e. Contract Abilities** are of 2 types:
 - **When Fulfilling.** These Abilities occur once only, as the contract is fulfilled.
 - **Fulfilled Ability.** These Abilities remain active after the contract is fulfilled and becomes an *enterprise (4B8)*.

HUMAN ICON [4B2d]

TOPHAT ICON [4B2f]

EXAMPLE [4B2e]

You win the auction for the Exodus contract “Rival Isobank”. You boost it to your Home Bernal, and take it and 5 isotopes to the listed destination, an opponent’s Bernal at Titan. Because of its “when fulfilling” Ability, you gain 2 isotopes at this Bernal, and 1 more at the isobank. Having fulfilled the contract, you gain the fulfilled Ability, which is to swap 1 isotope for 6 Aqua at your (soon-to-be-built) non-Home Bernal.

- f. Tophat / Felony Stack Abilities** (these 2 Abilities are active before fulfillment):
 - **Tophat.** A Stack containing a contract with the tophat icon can safely enter 1 Hazard per Turn without paying FINAO (the costs are thematically paid for by the billionaire).
 - **Felony.** A Stack containing a contract with the Felony icon ☹️ is allowed to commit Felonies (see glossary).
- g. Insight / Bailout “Fulfilled” Abilities:**
 - **Insight.** After announcing a *research auction (I2)*, you may initially choose one patent deck and move its top card to the bottom of its deck. Then choose any top card to auction. You may perform this Ability once per Turn, even if you have Insight from multiple sources. It is inactive during *research grants (O5e)*.
 - **Bailout.** This Ability grants immunity from *budget cuts (K2f)*. This is *negotiable (N7)*.

FELONY ICON [4B2f]



should be superimposed over the Humans on Board triangle

EXAMPLE [4B2g]

You gain the insight ability after fulfilling the contract “Hexapod Explorers”. You initiate a research auction, and apply insight to the generator deck, hoping to reveal a pulse generator. But alas, the revealed generator is even worse. You are obliged to research something, so you put a Freighter up for auction.

EASILY MISSED: You can negotiate a late contract. The payments transfer with the card. [4B3d]

TIP: Due to late fees, bidding in the first solar cycle is risky unless you boost a rocket quickly. [4B3]

EASILY MISSED: There are no contracts which match Centrist. If the Active Law is Centrist, none of the contract decks may be selected except (post-isobank) the Exodus contract deck. [4B3c]

TIP: Freedom/Individuality is almost all-manned (11 of 12); Equality/Authority is somewhat manned (4 of 12); and Honor/Unity is almost all unmanned (1 of 12 manned, 2 of 12 on the gold side). [4B3c]

h. Negotiation. You may trade your unfulfilled contract as part of a *negotiation* (N) replacing your timepiece with your partner's timepiece and transferring any late fee Aqua on the contract. The contract card may be transferred via *cargo transfer* (G1), etc.

- **Contract Abilities** are not negotiable unless specified.

i. Fulfilled Contract VP. See 4E2a.

j. Unfulfilled Contract Penalty. An unfulfilled contract suffers a VP penalty equal to how much you still owe (4E2c).

k. Cybernetics. These icons are active after a contract is fulfilled and becomes an *enterprise* (4B8). They enable the *implant* of the corresponding augmentation chit (4D4b).

4B3. Contract Auction (adds to the Fundraise Operation)

At the final step of a *fundraise operation* (O2b), if you do not have an active timepiece for another contract in the Sunspot Cycle, you may perform a contract auction for the top card in a contract deck sharing an Ideology with an Active Law. Before choosing to start the auction, you may Discard the top card from that deck.

a. Partner Enterprise. Whether or not there is a contract auction, a delegate from a fundraiser can be used to form a *partner enterprise* per 4B8a instead of being placed in the Political Assembly.

b. Auction Prohibition. Contract auctions cannot be held during War/Anarchy.

c. Deck Selection. After performing the fundraiser and associated *vote tally* (O2a), choose a contract deck sharing an ideology in its *ideology pair* (4A3e) with an activated law.

- If the Active Law is in Unity, multiple laws may be activated; otherwise, only one law will be activated.

- Once the *isobank* is built (4C2), you may also choose the Exodus contract deck regardless of Active Law.

EXAMPLE [4B3c]

You perform a fundraiser while the Active Law is freedom. You add a delegate into equality and move another delegate into equality so that freedom, equality, and unity each have 2 delegates. You examine the contracts, and the honor-unity ideology pair is the best one. Therefore, to break the tied vote tally, you move the Active Law to unity so that you may choose the top honor-unity contract to auction.

d. Proposal Review. You may choose to Discard the top card from the chosen deck.

e. Auction Start. Choose to either initiate a contract auction for the top card or abandon the auction.

f. Bid Media. Bids are made in **years**, from where the Sunspot Cube is currently located. Bids must start at 10 years (highest) and can go as low as 0 years (lowest).

- g. Auction Procedure.** If you initiate this auction, you must start the bid at 10 years. Then, in any order, players (with an available timepiece of their color) may bid. Continue until no player wishes to lower their bid. The lowest bid is the winner (minimum bid of 0 years).

EXAMPLE [4B3g] Player Purple places the Search For Life Center contract up for auction, automatically bidding 10 years. Player Red has a contract; their timepiece is in the Sunspot Cycle and so is not available for bidding. Player Green bids 9 years; Player Purple responds with 9 (winning ties as the active player). Bidding goes down to 3 years until Player Green passes. Player Red is committed to founding the “Search For Life Center” in 3 years’ time.

- h. Ties.** As the initiator, the active player wins ties they are involved in. If they are not involved, they choose which of the tied players wins. The Marketeer faction privilege, if active, always wins ties.
- i. 3-step Win Procedure:**
- **Deadline.** The winner places their timepiece (“sand timer” faceup) into the Sunspot Cycle a number of steps into the future equal to their bid.
 - **Cash Up Front.** The winner takes the *cash advance* (4A3h) in Aqua from the Pool.
 - **Hand.** The winner takes the contract as a Hand Card. It ignores *academia hand limits* or *budget cuts* (4B2a,c).
- j. Finder’s Fee.** If the auction is won by a player other than the active player, the active player may perform a second Operation on their Turn, which cannot be a *fundraise operation* (O2b).
- k. Postwar Space Empire (Module 3).** If the political assembly has been dissolved (e.g. if a *Space Empire* (3F5b) is formed), shuffle all the contracts (except *Exodus* contracts) into a single deck instead of three. This single deck can be chosen under any Active Law, including Centrist.

TIP: Contracts will be put into a Stack on the map via the boost operation (I4). Secure a contract before boosting your rocket; that way you can boost both in a single operation. [4B3i]

4B4. Contract Fulfillment (new free action)

Each step of a contract’s destination is fulfilled as a *free action* (G) by moving your Stack containing the contract to the listed destination, while satisfying any other listed conditions. If you fulfill the terms of the contract, move it to a space adjacent to your playmat, and return your timepiece from the Sunspot Cycle placard, which instantly lifts restrictions on contract auctions. Any “Fulfilled:” Abilities of the contract are now active (4B2e).

- a. Contract Start.** All contracts are put onto the map via the boost operation, boosted to your choice of either LEO or your Home Bernal, see 4A7a). If the contract specifies “Return”, it must be returned to your Bank.

EASILY MISSED: A contract cannot be voluntarily Decommissioned/Discarded. [4B4b]

EASILY MISSED: If you have an ET Home Bernal (Module 3), this is a valid return location for contracts. However, because you cannot boost there, if you have an ET Home Bernal you may only pick up contracts in LEO. [4B4d]

EASILY MISSED: As the Glitch Trigger occurs before the free action, it could destroy the contract or a piece of needed equipment before the contract can be fulfilled. [4B4f]

b. Representation. The contract card itself must be included in the Stack which fulfills the contract or destination step.

EXAMPLE [4B4b] A player is completing the “rivalry” contract (visit a Site in a zone with a claimed glory chit & return). She lands in the E Zone, which has a claimed glory chit, and announces this. On a future Turn, she realizes that the F Zone chit is a better target, so travels to a F Zone Site and announces the landing. The rivalry for the E Zone chit is no longer relevant.

c. PR. You must announce each step of a contract’s fulfillment to all players. Only the most recent announcement for each step is valid.

EXAMPLE [4B4c] The “Life Support Test Protocol” contract has a destination: “Site in Saturn Zone with Promoted Bernal and augmented Colonist & Return”. To fulfill this contract, you must have the contract, a Promoted Bernal, and an augmented Colonist at a Saturn Site, then must have those same cards at your Bank (the “Return” clause).

d. Return. If your destination specifies “Return,” after reaching the other destinations listed, the Stack must move to your *Bank (4A7a)* to fulfill.

- **Round Trip.** If all other destinations on a contract require cards to be with the contract, the “return” destination also requires the same cards.

EXAMPLE [4B4d] The tourist contract “Inner Tour” has a destination “Pass through 4 different flyby spaces in 1 Turn.” Your spacecraft, with the contract aboard, navigates the Earth, Luna, Mercury, and Venus (requiring season blue) flybys in a single Turn. Because the contract says “Pass through,” you do not need to stop at the final flyby, but you must fulfill the contract on this Turn, or you will have to do the tour again!

e. Presence. The contract and any other required cards must be at the listed destination or step.

- **Pass Through.** If the destination or step includes “Pass through,” the Stack instead must have passed through the listed location this Turn.

f. Glitch Trigger. Contract fulfillment (including intermediate steps) is a Glitch Trigger.

g. Contract Added to Your Contracts Fulfilled Area. If this is the last (or only) destination, the contract is fulfilled. Orient/splay the card together with your other fulfilled contracts next to your playmat so that its *cybernetics (4A3f)* and *fulfilled ability* (if any, **4A3d**) are visible. Return your timepiece to your Reserves and Discard any Aqua from late fees on the contract.

- **Abilities.** Any “When Fulfilling:” abilities take place at this point (after removing the contract from its Stack).
- **“Colocated”** refers to the contract location at the start of fulfillment (even though the contract is no longer there).
- **“Fulfilled”:** These Abilities are now active.

4B5. Late Penalties

If the 1st player cube moves *past* your timepiece in the Sunspot Cycle, you are **late** which flips the timepiece and orients the contract sideways. You can also declare a contract late, to pay it off faster. Pay its annual late fee until either it is fulfilled or its advance is paid (or twice the advance if the contract was never boosted). Follow these steps:

- a. **Flip To Deadbeat.** Flip your timepiece to its “deadbeat” side. It remains in place until removed by *fulfillment* or *repayment* (bullets d or e).
- b. **Pay Late Fee (optional).** At the start of your Turn in which your timepiece is “deadbeat”, *after Bernal profits*, you may pay all or part of the **late fee** listed on the contract. Place all payments directly upon the late contract to track how much you have spent.
 - **No Overpayments.** You cannot pay more than the late fee each Turn.
 - **Nonpayment.** Until the debt is paid, if you fail to pay the late fee then you *default* (4B6), receiving the default chit and moving the contract to your Hand if not already there.
- c. **Orient Card Horizontally.** A contract in horizontal (landscape) orientation indicates that it is late. This can be in your Hand, or in a Stack.
 - **All Abilities** (including *tophat* or *felony*, see 4B2f) on a late contract in are deactivated.
 - **Boosting/Decommission.** A late contract can be Decommissioned to your Hand, without changing the landscape orientation of its late status or the payment on it. A late contract cannot be boosted.
 - **Mishap Insurance.** If the Mishap Insurance Law is active when your timepiece flips to deadbeat, place Aqua from the Pool equal to the contract’s **Mass (4A3b)** on the contract. You may *lobby* (O4) this Law.
- d. **Late Fulfillment.** The instant a late contract is *fulfilled*, see 4B4g. This Discards the payments and timepiece.
- e. **Debt Repayment.** The instant the payments on the card are equal to its **cash advance (4A3h)**, Discard the contract with its payments and remove your timepiece from the Sunspot Cycle. If the contract was never boosted, then the payments must equal twice the cash advance. See the example below:⁷

"DEADBEAT" TIMEPIECE
[4B5a]

EASILY MISSED: Defaulting has no effect if you already hold the default chit and the contract is already in your Hand. [4B5b]

⁷MERCURY, NOT FOR BEGINNERS. Why is the closest planet to Sol one of the hardest and highest delta-v planets to reach? First, it orbits so quickly (about 45 km/sec) and so deeply in the Sol gravity field, inclined at 7 degrees from the ecliptic. Furthermore, without an atmosphere, aerobrake is not an option. NASA was unable to attain a satellite orbit around Mercury until the MESSENGER Spacecraft of 2011, which needed 6 years and flyby of Earth, Venus, and Mercury. The programmer responsible was Robert Farquhar, a planetary trajectory pioneer who designed some of the most esoteric and complex spacecraft trajectories ever attempted.

EASILY MISSED: If your contract blows up (perhaps an Act of God or a military action), then it goes into your Hand. This means that you have a second chance to boost it. [4B6a]

TIP: Defaulting on a contract, especially a manned one, can be very, very bad. Doing any of the following is like landing on Sol and hoping a sunspot will save you [4B6]:

- Make a Belt Roll which could Decommission a contract.
- Make a Hazard Roll which could Decommission a contract.
- Leave a Stack in space before a potential CME Event that could Decommission a contract.

REMINDER (MODULE 3): An opponent is another player, a player that is not you, the player whose Turn it is. During War, an enemy is an opponent who is in the bloc you are fighting against. [4B6a]

EXAMPLE [4B5]

You won the bid for the “Closest Planet” contract, and accepted its 13 Aqua advance, but getting to Mercury is harder than you thought. The 1st player cube moves past your timepiece (1). Flip your timepiece to the deadbeat side (2). You realize that it is impossible to complete the contract within the debt period, but wish to pay it off so as to avoid the default chit. On your Turn, you pay the late fee of 4 Aqua placed on the contract (3). On your next 2 Turns, you pay the late fee of 4 Aqua each Turn. On the Turn after that, you pay the last Aqua (4). Now that 13 Aqua are on the contract, it is Discarded along with its payments (5), and your timepiece is removed from the Sunspot Cycle (6). This frees you to be able to bid on more contracts.

4B6. Default Penalties

If unable or unwilling to pay the late fee, or your *manned contract* (4B2d) suffers an involuntary Decommission, you **default** and suffer the default chit.⁸

- Add to Hand.** Move the contract to your Hand. If it is late, it is added to your Hand in a horizontal Orientation to indicate that you must pay late fees in order to Discard the contract (4B5b,e).
- Take Default Chit.** Take the *default chit* (4A2g) from anywhere, and store it on your playmat on its blue or gold side depending on the contract color.
- Default Stigma.** If you have the default chit you are not allowed to boost manned contracts.
- Endgame Default Penalty.** During *endgame scoring* (M2, 4F), the default chit subtracts 3 or 6 VP.

EXAMPLE [4B6]

You boost a tourist contract, but it perishes in a solar flare with still 7 years remaining. The contract goes into your Hand, and you get the default chit. You reboost the contract (presumably with more naive tourists), and fly towards your destination. Before you reach it, you become late and start paying late fees. But finally you fulfill the contract and add it to your contracts fulfilled area. Meanwhile, your opponent also defaults on a contract, so you fortunately lose the default chit and its stigma.

4B7. Contracts & War/Anarchy (Module 3)

With the Conflict Module, contract auctions and fulfillment are slightly different during War / Anarchy: military contracts enjoy special importance, late fees are suspended, and contracts may be abandoned without penalty.

- War Outbreak Partner Dissolution.** If War occurs, or if during War an ally becomes an enemy, all partner delegates (4B8b) on enemy enterprises are Discarded. See 4B8c for re-installing during War.

⁸ DEFAULT is the failure to fulfil a contractual obligation. It damages reputation, the most fundamental asset of any enterprise.

- b. Military Contracts.** Contracts of type “military” remain unaffected by War/Anarchy, and ignore all the bullets below.
- c. Auction Freeze.** No contract auctions (except military) are allowed during War/Anarchy.
- d. Late Fees.** No late fees are paid (except military) during War/Anarchy.
- e. Abandonment.** A contract (except military) may be abandoned (without penalty) during War/Anarchy by taking its card to LEO or the Anchored Bernal of any player. Discard the contract and remove your timepiece from the Sunspot Cycle.

EXAMPLE [4B6e] Anarchy occurs in Season Blue, and you worry that an opponent will attack your Rocket Stack carrying a billionaire contract to cause you to default. You dump the billionaire at a nearby Bernal.

4B8. Enterprises (Modules 1 & 2)

A completed contract stored adjacent to your playmat is called an **enterprise**. The different types of cybernetics icons on them are important for implanting *augmentations* (4D3).

- a. Delegates & Partner Enterprises.** During a *fundraise operation* (O2b), you may place a delegate on an opponent’s enterprise (see next bullet). An enterprise with your delegate is called a **partner enterprise**.
- b. Placing Partner Delegates.** You may place a partner on an enterprise in an opponent’s enterprise only with their negotiated permission.
- c. Placing Partner Delegates (Module 3).** During War, you cannot place or retain partners on enemy enterprises, but you can place on allied enterprises, even without permission, as long as you pay the owner a *bribe* (3B1).
- d. Portfolio.** The cybernetics icons on your enterprises or your partner enterprises is called your portfolio. This is important for the *implant operation* (4D4) and thus for the *promotion operation* (4D7).

EXAMPLE [4B7d] You wish to promote your muddling GW thruster into a spectacular TW thruster, but only have 3 of the 4 implantations needed. Player Gray has a Jars cybernetics that you lack. During a fundraise, you place your Delegate on his fulfilled Contract. Then on the following turn, you move your thrice-augmented Colonist to the Home Bernal of Player Gray to perform the 4th implantation as a negotiated on-site operation (N6). In return for these services, you transfer ownership of your second Bernal to become his second Bernal.

4C. Isobank (Module 1)

The ready availability of Aqua to Earth and the mining of rare isotope fuels causes the Aqua market to crash, making isotopes the new standard. Inflation causes FINAO costs in Aqua to double, and the exchange rate of Aqua to Isotopes is 10:1. Eventually a new storage facility for isotopes, the **isobank**, becomes the new center for storing and exchanging isotopes.

TIP: The Isotope monetization cards included with Module 4 may be handed out as a reminder of the effects of monetization. [4C1]

4C1. Isotope Monetization (same as Module 3)

In Module 1, gold beads representing starship isotope fuel can be *factory-refueled* (**1C1a**) at factories matching the Spectral Type of your GW thruster, should you have one. This Spectral Type is called your **iso-standard**. In Module 4, the first player who establishes their iso-standard by producing a gold bead at a Factory triggers isotope monetization.⁹ This doubles the costs of FINAO for all players. Gold beads stored at your Bank can be used as currency for auctions or FINAO. Gold beads at your Bank are worth 10 Aqua.

- a. 2X FINAO Costs. Failure Is Not An Option (H7e)** costs are instantly and permanently doubled for all players (including those with Open Source faction privilege) if one of these occurs:
- Either the *Lofstrom Loop* (**2B4f**) or the *GEO Space Elevator* (**2B4i**) is Anchored as a Home Bernal.
 - The first gold bead of isotope fuel is produced (**1C1a**).
 - (Module 3 only) After a War, the first *ET Home Bernal* (**3F1**) is Anchored.
- b. Blue/Gold Bead Exchange Rate.** At your Bank, one gold bead can be exchanged for 10 Aqua, and vice versa. Thus, if you pay the programmers with 1 gold bead, you get 2 Aqua in return.

EXAMPLE [4C2b]

You have a mini-mag orion that you wish to take through a Hazard and need to pay the programmers. You isotope refuel one gold bead at your M Factory, and transport this by Freightier to the Bernal that is your Bank. Then you pay the programmers 1 gold bead, and receive 2 Aqua in change.

- c. Gold Bead Bank Transactions.** If your Bank receives any gold beads in a transaction, either through negotiation or in an auction, as soon as they leave the Bank they automatically become your iso-standard.

⁹ISOTOPE MONETIZATION describes the crash of the Aqua market, as a result of LEO becoming eclipsed as a space hub, supplanted by Bernals in water-rich areas. The new commodity economy is based upon isotopes, the sort of material needed for clean nuclear reactors for both Earth and starships, who are perhaps equally thirsty for energy.

- d. Debasement.** If you never had a GW/TW thruster and thus no iso-standard, you cannot hold any gold beads. If your Bank receives an Isotope in trade, you must immediately convert it into 10 Aqua.
 - If you receive an Isotope not at your Bank, you must immediately convert it into 1 FT locally.
- e. Iso-Standard Permanency.** See **1Cb**.
- f. Exchange Timing.** You may only exchange Aqua/gold beads as a free action on your Turn or when making a payment, excluding *negotiations* (**N**).

4C2. Founding The Isobank (new in Module 4)

An isobank is the Fort Knox of the Solar System, a Bernal where all the Solar System's reserves of Isotope Fuel (see glossary) are stored. This central repository is of military significance, and will eventually become the drydock for the interstellar rocket called Project *Exodus*. The player owning the isobank Bernal is known as the isobanker.

- a. Founding Bernal.** You may, as a free action, establish the isobank (and Project Exodus) if, on your turn, there is no isobank and you fulfill these 2 conditions:
 - Have an iso-standard (i.e. ET produced a GW thruster per **1Cb**).
 - Have an Anchored non-Home Bernal with a dirtside matching their iso-standard. (This can be at an opponent's Dirtside). If using Module 3, an ET Home Bernal can be the isobank.
- b. Place Starship Token.** Place the starship token on the map under the Bernal's token to show that it is the isobank. See **4C3a** if the Bernal is unanchored (voluntarily or involuntarily).
- c. Exodus Contracts Active.** Removing the starship token from the *Exodus* contract deck activates it; this deck is now available for *contract auctions* (**4B3**), and is now affected by inspiration events.
- d. Storage of Isofuels.** Once established, the isobank becomes the Bank for all players' isotopes.
 - **Isotope Stacks.** Each player places an isotope monetization/isobank stack card next to their playmat, isobank stack side up. The player founding the isobank takes the single Isobanker card. This card represents a new Stack located at the isobank.
 - **Isotope Movement.** When the isobank is established, isotopes at player's Banks move to their isobank Stack. If the isobank moves, isotopes at the isobank move with it.
 - **Exchange of isotopes** is to/from the isobank rather than to/from LEO/Home Bernal.
 - The isobank does not count as a Bank for any other purpose (return of contracts, exomigration, etc.).

EASILY MISSED: If you Anchor a Bernal to an opponent's Dirtside, you do not automatically get permission to use it. [**4C2a**]

EASILY MISSED: After the isobank is built, gold beads at LEO or your Home Bernal/ET Home Bernal cannot be exchanged to Aqua - you must take them to the isobank. [**4C2d**]

- e. **Hoarder.** The isobanker cannot sell isotope for Aqua (but they can buy isotope with Aqua). Other players can sell for them, but may of course negotiate a fee.
- f. **Investor's Capital.** *Isotope refuel (1C1a)* at the isobank or its Dirtside is doubled.
- g. **Founding Bernal Loss.** If the isobank's Bernal is no longer Anchored and/or owned by the isobanker, (e.g. it was unanchored voluntarily or in combat), the isobank/starship token automatically moves to one of its Dirtside (isobanker's choice). The next player to meet the isobank requirements (at this or another location) becomes the new Founding Bernal and the new isobanker. All isotopes in the isobank are now at the new location.
 - **Takeover.** If a player Anchors a Bernal at an isobank whose Founding Bernal has been lost or unanchored, they may become the isobanker without meeting the requirements of **4C2a**. The isobank remains at its present location.

EXAMPLE [4C2g]

Player Green Anchors a Bernal over badenia, where they have a Factory and have previously mined isotope (setting their iso-standard to D). They choose to establish the isobank. Their isotope is at an outpost on badenia, so does not move. Player Red has 3 isotopes on their playmat, thus at their Bank. Player Red's isotope remains on their playmat, but its location is now the isobank. Player Red chooses to sell 1 isotope. They remove 1 isotope from their playmat and place 10 Aqua there. The Aqua on their playmat is at their Bank (LEO or Home Bernal).

- h. **Iso-standard Change.** The isobank is not affected if the iso-standard of the founder changes.

EXAMPLE [4C2h]

You ET produce an "Amat-catalyzed Fusion-Fusion" GW thruster, of Spectral Type S. Your second Bernal, anchored at helene of Spectral Type S, becomes the isobank. Place the starship token under your helene Bernal. Your opponent also ET produces a GW thruster, but since your opponent's isotopes are now stored at the helene Bernal, he must move his GW thruster there in order to fuel it from his Bank.

4C3. The Isobank/Starship During War or Anarchy (Module 3)

The isobank stores the isotopes of all the players, both allies and enemies. If the founding Bernal is unanchored, the isobank is sent Dirtside for protection (**4C2g**) where it remains in space), where it can be destroyed or captured in combat. During or just after War/Anarchy (assuming that it has enough isofuel, crew, and equipment), the isobank can launch as a **starship** and attempt to fly off the map, with the player with the most isofuel in command. Exiting the map prematurely ends the game with scoring. Players may attempt to stop this by combat per **4C5c**.

- a. **Isobank Protection.** During War or Anarchy, the starship token automatically moves to a Dirtside (isobanker's choice) of the Founding Bernal. It remains there during War / Anarchy regardless of Founding Bernal Loss (4C2g). When War/Anarchy ends, the isobank returns to a Bernal if there is one using its location as a Dirtside.
- b. **Destruction & Bank Heist.** The starship token itself is unaffected by combat. If at a Factory which is destroyed or captured in combat, the victorious player steals the stored isotopes of all the other players, in the heist of the millennium. Then, the isobank is destroyed, the starship token is discarded, and a new isobank may be established per 4C2. Place the isotopes from all players' playmats in an outpost at the former isobank location.
- c. **Desperation Starship Exodus.** During War, or during the year the War ends, the player with the most isotopes in the isobank can convert the isobank into the Starship *Exodus* with themselves as commander. They remain in command for as long as the starship exists. (If tied for the most isotope, the isobanker chooses whether to allow exodus.) This option requires the following requirements at the isobank:
 - **TW Thruster.** The starship needs an operational TW Thruster at the isobank.
 - **Augmented Colonist.** The starship needs at least one augmented Colonist.
 - **Fuel.** On departure and upon exiting the map, the combined number of isotopes must be at a minimum "reckless", see 4C4b. The commander robs the isobank of the isotopes from every player for his getaway.
- d. **Exodus Launch.** As a free action the commander moves all the starship components (previous bullet) into the Starship Stack placard (4A2c). If the isobank is on a Site, the starship launch is allowed an automatic risk-free factory-assist.
 - **Exodus Cargo** can only include the following Promoted cards: (1 TW Thruster, 1 Freighter, 1 Bernal, and 1 or more augmented Colonists). Supports are not required. The TW Thruster and at least one augmented Colonist are mandatory. Any amount of Isofuel can be carried.
 - **Crew Selection.** Any augmented Colonist present at launch may decide to join the starship. Award "passenger" Exodus medal (4A2n) to each passenger.
 - **Isobank Stacks.** Isobank Stack cards are flipped to their isotope monetization side; any Stack contents are converted to Outposts.

TIP: Submarines make impregnable isobanks. [4C3b]

EXAMPLE [4C3d] War has broken out in a 3-player game, and Player Yellow, fighting for the Loyalist bloc, has the most isotopes (10). He negotiates with the enemy, Player Gray fighting for the Independence bloc, to send a TW thruster with an augmented Colonist to the isobank at Titan, and converts it into the Starship Exodus. This steals 3 isotopes from the other players, so that Exodus flies with a reckless 13 isotopes.

TIP: Freighters have a net thrust of 1, are pushable with Powersat, and do not track fuel consumption. If a Promoted Freighter is part of the desperation starship, the commander's Mobile Factories retain the characteristics of that Freighter card. [4C3e]

EASILY MISSED: Other spacecraft cannot stop in the solar oberth space, so only ranged/kamikaze weapons can be used to attack the starship there. [4C3g]

- e. **Exodus Movement.** The commander flies the isobank/starship token as their Starship Stack using freighter stack movement (1B4), with 3 Bonus Pivots.
 - **Towing.** A fueled starship has a mass of 220 (max) or 80 (reckless), so all starships are far too heavy to be towed.
 - **Cargo Lock.** Once underway, the starship cannot change its Cargo except to add additional isotopes or augmented Colonists by cargo transfer.
- f. **Exodus Anarchy.** Launching the starship causes Anarchy, which lasts until the starship exits or is destroyed.
 - **Flag of Truce.** Combat in the starship's space is always optional. Cards selected for the starship cannot fire, even if they are enemies.¹⁰
 - **Combat.** The starship token has Rad Hardness 10 (thus has a gun in combat).
 - **Destruction.** The starship is destroyed if all contents are Decommissioned (via a Crash Hazard, for example), or if neither Colonists or TW Thruster remain aboard. The entire starship stack including isotope fuel is Decommissioned.
- g. **Exodus Endgame.** If the commander uses one of the three exits to leave the map, the game ends at the end of the Turn, with scoring per 3G1 and 4E2, including Project Exodus VP for those aboard the starship per 4E3.
 - **Starshot.** *Project Exodus scoring (4E3)* is used, using only the fuel, equipment and crew on the exited starship.
 - **Jupiter Exception.** If the Jupiter-Sol-Jupiter exit is used, move the starship to the *solar oberth (H8e)* (but don't apply its crash Hazard since you have already done it in the exit path), where it remains for two Turns (and can be range or kamikaze attacked, similar to dropstones). At the end of the second Turn, if the starship still has an operational TW Thruster and at least one crew, it exits and ends the game.¹¹

EXAMPLE [4C3g]

Because of the chance of enemy intervention, the desperation starship from the previous example does not dare to take the Jupiter-Sol-Jupiter exit, so it instead flies toward the Neptune Oberth exit. In 5 years it exits the map and ends the game.

¹⁰ STARSHIP EXODUS. Although its power is in the terawatt range, and a thrust of around 9 in the High Frontier scale, its enormous mass gives it a rather sluggish acceleration. Its three ballerinas are based upon an estimated specific impulse of 2449 ks (8% c).

¹¹ JUPITER-SOL-JUPITER EXIT. This is the most fuel efficient route to exit the solar system, but leaves the starship in the inner solar system for a vulnerable two years. The initial flyby of Jupiter kills the ship's orbital velocity, allowing it to plummet towards Sol. This takes 1.1 years, and the engine is ignited at perihelion, when it is closest to Sol. The resulting Oberth force multiplier and continued thrust takes it clear of the solar system, with a final pass at Jupiter again.

4C4. Exodus Fueling Table

Starships require a tremendous amount of Isotope Fuel (Isotofuel).¹² The minimum “reckless” amount of fuel required for a starshot to occur depends on player count as shown in the table below. Also shown is the starting fuel in a continuing *Interstellar* combined game. The player providing the most isotofuel chooses the starshot commander (4C3c).

- a. **Isototal** is total isotopes owned by all players collectively, in all locations.
 - Isotope present at the completion of *ad astra futures* (1D1b) counts toward that player’s isotope contribution towards Project *Exodus* (4E3b,d) even though it has been Decommissioned. Track this by storing gold beads on the Future card (4Agi).
- b. **Starshot Determination.** If the isototal at the endgame meets the minimum number in the “reckless” row for the appropriate column based on player count, there is a starshot and *Project Exodus scoring* (4E3) takes place.
- c. **Fueling Bonus or Penalty.** If there is a starshot, players with passengers receive an endgame VP bonus or penalty based on the isototal as shown in the final column of the Exodus Fueling Table (below).
- d. **Interstellar Fuel Levels (Combined Game).** Compare the isototal to the appropriate column based on player count to see the starting wet mass for *Interstellar*.

EXAMPLE [4C4]

During the endgame starshot, Player White has 10 isotopes, Player Yellow has 2 isotopes, and Player Green has 1 isotope plus 10 isotopes for completing the “Lithiated Ammonia Ice Starship” Future. The isototal is 23, which is “reckless” in a

3-Player game.

Starting Wet Mass in <i>Interstellar</i>	1 Player isototal	2 Players combined isototal	3 Players combined isototal	4 Players combined isototal	5 Players combined isototal	6 Players combined isototal	Endgame VP adjustment (per player)
Max (220 mass)	15 or more	24 or more	30 or more	40 or more	50 or more	60 or more	+10 VP
Crisis (130 mass)	10 - 14	15 - 23	24 - 29	32 - 39	40 - 49	48 - 59	0 VP
Reckless (80 mass)	5 - 9	7 - 14	12 - 23	16 - 31	20 - 39	24 - 47	-10 VP

¹² ISOTOPE FUEL. A typical flight plan for the Starship *Exodus* starts with a wet mass of 220 mass (8800 tonnes), of which 10 mass is dry mass (equipment and passengers), and 210 mass (210 tanks) is isotope fuel for the fusion engine. It expends 190 of those tanks accelerating to cruising speed (16%c) in 36 years, leaving 20 mass of braking fuel. Each tank is half helium-3 and half deuterium isotopes, refined from the atmosphere of Saturn, Uranus, or Neptune.

4D. Spaceborns & Cybernetic Augmentations

Human Colonists come in two types: **earthborns** (comfortable in 1 gee planets and Bernal's) and **spaceborns** (Human-machine hybrids comfortable in zero-gee and high radiation). Colonists of all types become augmented if they are implanted with an augmentation. **Augmentations** are chips that represent technological modifications for humans, and biological modifications for Robots.¹³ They are placed on a Colonist card via the *implant operation* (4D4) to enhance the Colonist's capabilities.

EASILY MISSED: Earthborn Colonists are Human (and not Robots). Spaceborn Colonists are Human (and not Robots). Unemancipated Robots are not Human and not spaceborn (but are drawn from the spaceborn queue). Emancipated Robots are Human and spaceborn. [4D1]

4D1. Spaceborn Colonists

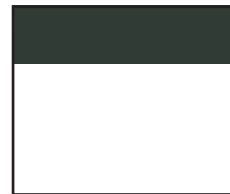
Module 4 introduces new Colonist cards called **spaceborns**, cyborg Humans whose bodies have atrophied so severely in the space environment that they are inseparable from their augmented limbs, locomotion, and life support system. Unlike earthborns (Colonists introduced in Module 2), spaceborns have artificial blood and are cancer-immune. Although they are Human (see glossary), they support Robot equality during *emancipation* (2A6b).

- a. **Colonist Provenance.** Colonists may be spaceborns, earthborns, or Robots.
 - **Emancipated Robots (2A6b)** count as Humans, Robots, and spaceborns.
 - **"Colonist"** refers to any of the above.
- b. **Sterile.** Spaceborns and Robots cannot obtain the heart (parenthood) augmentation.

¹³ PROMOTION VS AUGMENTATION. Many of the promotions on Colonist cards are a combination of genetic and bionic enhancements, as well as the cybernetics that communicate with them. An extreme case is a "reverse cyborg", which has mental functions largely undertaken by an exocortex. On an interstellar mission, augmentations of human biology are essential because the expected radiation levels are not survivable for pure humans. In the next game in the *High Frontier* saga - *Interstellar* - the humans have further diverged into specialized cyborgs, Amazons with reproductive capacity and heavily-augmented brains, and Meatheads with hibernation capacity and who have heavily augmented bodies.

4D2. Spaceborn Exomigration (new option for exomigration free action)

This Module plays with 2 *queues* (2A2b): one for the Human Colonists of Module 2 (called **earthborns**), and the other for all Human Colonists from this Module (called **spaceborns**) plus Robots from either Module. During *exomigration* (2A6), you may choose which queue to draw from. However, to draw from the Robot/spaceborn queue you must have at least 1 Colony dome that is not your Home Bernal and must join the Spacer Union. Except per 4D2e or 4F2f, no player may hold Colonists from both queues.



- a. **Space-Based.** You may not exomigrate from the spaceborn queue unless since the beginning of the Turn you owned a Colony dome other than your Home Bernal.

EXAMPLE [4D2a] You have a Home Bernal anchored, and then industrialize your first Factory. You use your Colonist to make it into a Colony, and then exomigrate a new Colonist. This new Colonist cannot be spaceborn, because this Colony is your first outside your Bernal, and did not exist since the start of the Turn (too soon to produce spaceborn).

- b. **Spaceborn Exomigration.** Draw the top card of the spaceborn queue, following normal exomigration rules.
 - If Multiple Robots are Drawn, choose one per 4A6i.
- c. **Joining Spacer Union.** If you put a spaceborn into play, immediately Discard (not a Felony) all your earthborns in play and exomigrate, drawing from the spaceborn queue.
- d. **Earthborn Exomigration.** If you hold any spaceborn colonists, you are a member of the Spacer Union and are not allowed to exomigrate from the earthborn queue.
- e. **Rank & File.** If the spaceborn queue is empty or contains only Robots, your exomigration emancipates the Robots per 2A6b. If post-emancipation and the earthborn queue or the spaceborn queue is empty, players must exomigrate from the other queue (even if they do not meet the Space-Based requirement listed above). the Spacer Union is dissolved, so that 4D2c and 4D2d are ignored

EXAMPLE [4D2] You establish a Colony on Ceres, and then Anchor a Home Bernal. Your Ceres Colony allows you the option to exomigrate from the spaceborn queue. The first 3 cards that you draw are Robots, and finally you draw a Human which goes into your Bernal Stack. You examine the 3 Robots, choose one, and shuffle the 2 remaining back into the spaceborn queue.

EASILY MISSED:

Augmentations cannot be removed except by Decommissioning the Colonist. You also cannot exchange ownership of Colonists (N3d). [4D3]

4D3. Augmentation Chits

Augmentations are chits that come in 6 shapes, listed below. Augmentation chits are placed on Colonists via the *implant operation* (4D4), they are only removed from the Colonist during Decommission.

- a. **Isotope (triangle shape).** A Colocated factory refuel that produces isotope receives an additional isotope. Even if other Abilities increase isotope fueling (e.g. miner colonists), only a single extra isotope is provided.

EXAMPLE [4D3a] A factory refuel at a Site with a miner Colonist generates 2 isotopes (1, plus 1 for the colonist). If the "Femtochemistry" refinery were present, each of those would be doubled by the refinery's ability, for 4 isotopes overall. In either case, the Isotope augmentation would provide 1 extra isotope (3 total with just the Miner, 5 total with "Femtochemistry").

- b. **Battery (hexagon shape).** This provides a pulse generator (this is an on-board nuclear support, see 2C3).
- c. **Jars (jar-shape).** An industrialization that creates a Colony by Decommissioning this Colonist (even one without an ISRU platform) does not need to Decommission a robonaut (represents hibernation jars).
- d. **Life-Support (heart shape).** This enables a special implant operation called parenthood.
- **No Spaceborn.** This chit can only be placed on an earthborn.
- e. **Warden (rectangular).** This adds a buggy platform of ISRU = 1, see I6.1. This is a *buggy weapon* in War/Anarchy (3E4).
- f. **Shield (radiation triad shape).** This gives the Colonist a minimum rad-hardness of 9. This is a *gun weapon* in War/Anarchy (3E3b).

AUGMENTATION TABLE [4D3]

AUGMENTATION	ICON	EFFECT
Isotope		Colocated isotope refuel(s) receives +1 isotope
Battery		Pulse Generator
Jars		Colocated industrialization with colonization does not need a robonaut.
Life-Support		Parenthood
Shields		RH 9,
Wardens		ISRU 1 buggy

4D4. Implant (new Operation)

This Operation places one or more augmentation chits onto any number of Colocated Colonists (earthborn, spaceborn, and/or Robot) at an Anchored Bernal. The chits have **cybernetics icons** (engine, batteries, jars, life-support, wardens, shields); you must have corresponding icons on fulfilled contracts in order to implant. Additionally, you must have a Colony dome that is not your Home Bernal to represent the space-based conditions that lead to augmentation.

- a. Location Requirements.** This Operation must be performed at your Anchored Bernal (or an opponent's Bernal if negotiated or the Law is *freedom to roam* **O5f**).
- b. Space-Based.** You must have at least one Colony that is not your Home Bernal to perform this Operation. If playing with Module 3, your ET Home Bernal qualifies.
- c. Portfolio Requirements.** You must have cybernetics in your Portfolio with icons for each of the chits you wish to implant.
- d. Implantation Limits.** Each Faction is limited to one augmentation chit of each shape.
- e. Procedure.** Take one or more valid augmentation chits and place each of them on top of a Colonist.
- f. Parenthood.** If the life-support augmentation is one of those added, see **4D5a** for limits and **4D5b** for effects.

EXAMPLE [4D4]

You Anchor your "GEO Space Elevator" Home Bernal in GEO. Remember that this requires an Epic Hazard Roll (4A6k). This exomigrates the "Vatican Observers". Since you have a Colony on Ceres, you are qualified to implant the observers. The cybernetics on your fulfilled contracts include two hearts and a triangle, so you expend an Operation to implant the heart and the triangle chits, placed on the colonist card. The heart chit causes parenthood, see the next example.

4D5. Parenthood

The life-support (heart-shaped) augmentation can only be implanted in earthborn Humans, because the genetics of spaceborn suffer from their high radiation environment. This augmentation enables parenthood:

- a. Reproduction Requirements.** This chit can only be placed on an earthborn Colonist, and cannot be placed on Robots, spaceborn, or a Colonist with the Jars chit.
- b. Preteen Years.** When this chit is placed, place your child chit on the Solar Sunspot Cycle 9 years into the future. During this period, the child Colonist is considered part of the heart augmentation of its parent.

EASILY MISSED: Since parenthood Decommissions the parent first, you may choose that Colonist as the child (a "chip off the old block"). [4D5d]

c. Coming of Age. When the Sunspot Cube moves past the child chit, the child is ten years old. The life-support Colonist is Decommissioned, and replaced by an earthborn **(4D1)** Colonist of your choice. You may look at the entire Colonist deck, and choose the one you wish to be the newborn. Shuffle the deck after examining it. This does not cause any delegate placement.

d. Wunderkind. The child comes into play Promoted (Purple-Side)

EXAMPLE [4D5d] Your "Vatican Observers" from the previous example scandalously got pregnant in the last year of Season Blue. Place the child chit 9 years into the future, in the first year of the next Season Blue. When the Sunspot Cube moves into the 2nd year of Season Blue, the Vatican Observers are Decommissioned into their queue (perhaps the Pope was outraged). You choose the wunderkind "Malcolm" as the successor, promoted as "Renaissance Man".

e. Inheritance. The preteen inherits all the augmentations (including the heart) of the retired Colonist (its parent).

4D6. Augmented Colonists

An **Augmented Colonist** is a Colonist (Human or Robot) implanted with at least one augmentation chit.

a. Augmented Humans vs. Augmented Robots. Humans (either earthborns or spaceborns) can prevent or cause Felonies, collect glory, can repair Glitches, and it is a Felony to voluntarily Decommission them. Robots don't have these capacities unless *emancipated* **(4D1h)**. See Human and Robot in the glossary.

b. Promotion. If a Colonist is Promoted, move their augmentations onto the Colonist's Purple-Side.

c. Death. If a Colonist is Decommissioned (other than for parenthood), Discard their augmentations to their Reserves.

EXAMPLE [4D6c] During Anarchy, you decide to Murder/Suicide your Colonist left stranded on Mars after delivering a contract. The Colonist's Jars augmentation is returned to the Jars.

4D7. Promotion (Operation revised from Modules 1 & 2)

This Operation flips a stack card to its improved Purple-Side. **4d7a,b,c,d** are the same as Modules 1 and 2 (**1A5** & **2A3**), while **4D7e** is a new requirement unique to Module 4. In short, to perform the promotion operation, you must bring Colonist(s) with a certain number of unique augmentation chits as well as the card being promoted to either a promotion colony or a Lab Bernal.

- a. **Dome Icon.** This indicates the colony type where a promotion can take place. There are 5 types:
 - **Spectral Colony** if at the specified Spectral Type (**D, H, S**, etc.)
 - **Submarine Colony** (waves icon)
 - **Astrobiology Colony** (leaf icon)
 - **Atmospheric Colony** (cloud icon)
 - **Push Colony** (push icon)
- b. **Lab Promotion Site.** A Promoted and Anchored Bernal is always a valid promotion colony for Colonist, Freighter, and GW thruster cards (**2A3c**).
- c. **Negotiation.** To promote, the Colony or Lab must either be your color or be an opponent's color as a *negotiated on-site operation* (**N6**) or alternatively allowed by the *freedom to roam* law (**O5f**).
- d. **Decommissioning Promoted Cards.** Except for Robot sales (**2C2a**) and Robot transfers (**Nb**), Purple-Side cards cannot be sold on the free market (**I3b**) or included in a stack transfer (**N3**). They can be involuntarily Decommissioned, but see (**1B7**) and (**2C2**). They can be voluntarily Decommissioned (Felony for Human Colonists) and then sold or transferred from your Hand (except Human Colonists). They return to play on their White-Side or Black-Side by being boosted (Bernals), exomigrated (Human Colonists), or ET Produced (Robots, Freighters, and GW Thrusters).
- e. **Augmented Colonists (new Promotion Requirement).** In order to promote, a Colonist (or Colonists) of yours must be present with a certain number of unique augmentation chits:
 - **To Self-Promote Your Colonist.** The Colonist being promoted needs at least 1 augmentation.
 - **To Promote Your Freighter.** The Colonist(s) doing the promotion needs at least 2 different augmentations.
 - **To Promote Your Bernal.** The Colonist(s) doing the promotion needs at least 3 different augmentations.
 - **To Promote Your GW Thruster to a TW Thruster.** The Colonist(s) doing the promotion needs at least 4 different augmentations. Additionally, you must either make a Hazard Roll or pay FINAO. If you fail the roll by rolling a "1", your Colonist(s) is Decommissioned.

EXAMPLE [4D7a,e] You have 2 Colonists, one with a rectangle and another with a triangle augmentation. This is enough to promote Freighter. Your “Fusion Fragment Sail” has the promotion colony M. Accordingly, you fly it to your Colony on Iutetia and spend a promotion operation to flip it to its Purple-Side.

4E. Module 4 Endgame

4E1. Module 4 Endgame

The game ends when the final Seniority Disk is taken, or (Module 3) if the starship token exits the map as per **4C3**.

4E2. Module 4 Endgame Scoring

Scoring is per the Core Game (**M2**), or **3G1** if using Module 3, but with these additions:

- a. Contract VP.** Each fulfilled contract scores VP as listed. (Module 3) Apply a multiplier as for bribe VP (**3G1a**).
- b. Default Chit.** You suffer negative VP if you hold the default chit (-3 VP blue side, -6 VP gold side).
- c. Unfulfilled Contracts.** If you have an unfulfilled contract, you suffer negative VP for each Aqua left unpaid towards the advance (or twice the advance if the contract was never boosted, see **4B5b,e**).

EXAMPLE [4E2c] You end the post-isobank game with an unfulfilled contract with a cash advance of 17 Aqua. Your opponent also has a default contract, with a cash advance of 21 and with 10 Aqua of payments on the card. You suffer a VP loss of 17, and she suffers a VP loss of 11.

- d. Isobanker (4C2)** receives 5 VP.
- e. Project Exodus Requirements.** A starshot occurs with Project Exodus scoring (**4E3**) if either of these endgame conditions are true:
 - **Ad Astra Future** was completed (**1D1b**). See **4A6i**.
 - **Endgame Equipment.** A TW Thruster and a high enough isototal (**4C4b**) exists.
- f. If Tied.** Resolve by most Aqua. Otherwise, share victory.

4E3. Project Exodus Scoring

At the end of the game, if the *starshot requirements* (4E2e) are met, there is a starshot with additional Project *Exodus* scoring. This collects all the isofuel (including ad astra isofuel per 4A6i) and Purple-Side cards that exist in the game to see if they contribute to the Starship *Exodus*. Players are awarded medals according to their contribution:

- a. **Isotope Exchange.** In reverse player order, players may exchange Aqua for isotope (or vice versa).
- b. **Ad Astra or Desperation Starshot (Module 3).** For these starshots, the medals and VP of the next 3 bullets are scored only if they are present at the starship location when it departs (4A6i or 4C3c). For Project *Exodus* scoring, the medals and VP are awarded no matter where the passengers and equipment are located.
- c. **Award Command Medal (6 VP).** The player with the most isotopes (including ad astra isofuel per 4A6i) receives the command medal. If tied, the isobanker chooses among those tied; determine randomly if there is no isobanker.
- d. **Project Exodus Fueling.** Compare the isototal (4C4a) to the appropriate column of **Table 4C4** based on player count. If the result is reckless (4C4b), subtract 10 VP from each player with one or more augmented Colonists. If the fueling is max, add 10 VP to each player with one or more augmented Colonists.

EXAMPLE [4E3d]

In a 5-player game the isototal is 50 isotopes. All players have one or more augmented Colonists, and all 5 add 10 VP to their score per Table 4C4.

4E4. Combined Game-Interstellar (6-7 cycles game only)

The next game in the grand saga is *Interstellar*, humankind's journey to the stars. This competitive-cooperative game is sold separately, and is a complete game in itself.¹⁴ No player may continue to the combined game of *Interstellar* unless there is a *starshot* (4E2e) and they have provided at least one *augmented colonist* (4D6).¹⁵

¹⁴ BIOS GRAND CAMPAIGN. The *Bios* Series and the *High Frontier* series of games string together for one long campaign encompassing everything. If one starts with the origins of life (*Bios:Genesis*), continues to life getting big and conquering the land (*Bios:Megafauna/Mesofauna*), continues to life getting conscious (*Bios:Origins*), and then into *High Frontier*, concluding with the *Interstellar* game is necessary to complete the cycle. This is because the starship brings with it the seeds to start microbial life afresh, on a distant planet, so that it is *Bios:Genesis* once again.

¹⁵ PROJECT *DAEDALUS* was a design study for an unmanned interstellar spacecraft that could carry a mass 10 payload to a nearby star in half a century. The starship used terawatt fusion engines using isotopes of helium-3 and deuterium as fuel. The game *Interstellar* uses a version of Project *Daedalus* is upgraded to be able to carry humans and able to actually halt at the target star system.

—A Bond & A Martin, *Project Daedalus - The Final Report on the BIS Starship Study*, 1978..

TIP: Consider a map exit if making an Ad Astra Future with that exit or by making the desperation starshot (4C3c). [4E4d]

- a. **Cyborgs** are augmented Colonists who join the starshot.
- b. **Summary Interstellar Perks (C8).** The following gives you advantages in *Interstellar*, either to maximize mission success, or maximize game victory. See **C8** in the *Interstellar* rulebook:
 - **Winning the Module 4 Game (C8b),** This awards the patron medallion and you choose who is the first mate.
 - **Contributing the Most Isofuel (C8c),** This awards the command medallion and you choose who is in command.
 - **Contributing a Promoted TW Thruster (C8d),** You start with the propulsion medallion.
 - **Having Augmentations (C8e),** These enable your passengers on *Exodus* to start augmented.
 - **Glory & Medals (C8f).** These are worth 1 VP each in *Interstellar*.
- c. **Select Equipment.** If there exists a Promoted Freighter, Promoted Bernal, and/or Promoted TW thruster anywhere, one of each are requisitioned for Project *Exodus*.
 - **Contributing a Promoted Freighter (C8g),** *Exodus* starts with a “skinny” wardens blueprint, which allows you an extra slot.
 - **Contributing a Promoted Bernal (C8h),** *Exodus* starts with a “skinny” life-support blueprint, which allows you an extra slot.
- d. **Map Exit (C8i,j).** Taking the Jupiter-Sol-Jupiter exit (or if no exit was made) starts *Exodus* at speed = 0, but taking other exits starts *Exodus* at speed = ½. Taking the Sol Oort exit also allows a free reconnaissance (**C8k**).
- e. **Interstellar Starship Wet Mass (C8l).** *Exodus* starts with a variable amount of fuel, depending on how much the player’s have jointly contributed per **4C4d**.¹⁶

EXAMPLE [4E3e] Players Purple, Green, and Player White all have Promoted Bernals. Player Purple also has the Daedalus TW thruster, which has the highest base thrust (11) and therefore chosen as the engine for *Exodus*. Player Purple thus holds the “propulsion” equipment medal, making him ineligible to hold the Bernal medal. Since Player White is trailing in overall VP, he decides to award Player White with the “Bernal” equipment medal.

¹⁶ **PAYLOAD COST.** A starship of mass 10 (400 tonnes) traveling at its game maximum (15% lightspeed) would have a kinetic energy of 18 trillion joules. Every kilogram of payload on the starship that is accelerated to this speed would need 150 kilograms of isofuel releasing 45 million joules. This is equal to the total global energy usage each month in 2020, mostly from gas, oil, and coal. Converted to power, the 2020 global power demand of 17 terawatts is much less than the power of a starship engine (50 TW for *Daedalus* 1st stage, 4 TW for the 2nd stage). To achieve a delta-v for a cruising speed of 15%c, and assuming *Daedalus* engines with an exit velocity of 3%c, the mass ratio equals 150. This means that if one astronaut was 10 kilos overweight, it would cost the expedition man-years of miners and refiners to produce the additional one and a half tonnes of pure isotope fuel.

4F. Exodus Scenarios (Module 3 competitive)

These wargame variants require Module 3 (Conflict). They assume a total war between two implacable foes, and the loser is motivated to exit the solar system entirely, and start over in a new system. Conversely the winners of the War are motivated to hinder the losers from building a starship and escaping.

- a. **Sides.** One side (as specified) are the Loyalists and the other are the Independents. However, as this is an ideological rather than racial war, you are free to join the enemy side using defection (3B3c).
- b. **Desperation Starshot.** If you are the commander, you may attempt to end the game early per 4C3c, except that you may also attempt it after the War if you lost the War (i.e. your extremist delegate was disgraced).
- c. **Winning the War.** See 3F4.
- d. **Winning the Game.** See 4E2, including contributions to Project *Exodus* (4E3).

4F1. Exodus Conventional War

This game plays with all the rules of Module 3, but with scoring per 4E2. *Desperation starshots* (4Fb) are allowed.

4F2. Exodus Union War

The game begins with Human factions per Module 3, but if War breaks out (from any of the causes listed below), the Robots become introspectively conscious and join the spaceborns in a fight for recognition as fully human. Players are assigned to a bloc depending upon the Colonists they hold. Follow these steps:

- a. **Indicate Bloc Allegiance.** Players with Robot/spaceborn Colonists (i.e. belonging to the Spacer Union (4D2)) start as Independence. Players with earthborn Colonists start as Loyalist. Players with no Colonists must join either bloc, choosing secretly and simultaneously. Flip your propaganda chit to independence or loyalist without losing your extremist. Propaganda chits with no Extremists are unaffected.

EXAMPLE [4F2a]

As Player Purple, you default as Loyalist when War breaks out. Yet you hold the “Gabonese Mining Consortium” Colonist and thus are a member of the Spacer Union, which indicates that you start as Independence. Flip the propaganda in authority to its Independence side, with your extremist on top.

EASILY MISSED: If your exomigration is limited by the lack of suitable Colonists, you must wait until one is available. See 3A3g. [4F2b]

- b. Desertion of Enemy Colonists (modifies 3C3f).** All earthborn Colonists are members of the Loyalist bloc, and all Robot/spaceborn Colonists are members of Independence, regardless of ideology. If at the end of your Turn a Colonist of yours supports the opposing bloc than what your Propaganda chit indicates, you must either:
 - **Evade** the “No Enemy Colonists” restriction (**3D5d** honor).
 - **Discard** this enemy Colonist to the bottom of its queue.
- c. War Outbreak.** There are 3 ways that War can occur:
 - **A War Outbreak Roll** is made during each adjustment of the exploitation track per **3C1** to see if War occurs.
 - **Empty Queue.** War also occurs if there is an *exomigration* (**2A6**) but there are no cards in the appropriate queue to draw from.
 - **Uplift Future.** War also occurs if the *uplift future* (**1D5n**) has been completed. This player alone becomes the independence bloc.
- d. Stillborn War.** If War breaks out with all the factions and all the extremists in the same bloc, then the war ends without any War effect (including no Space Lord effects).
- e. Exomigration.** Independence can only *exomigrate* (**2A6**) from the Robot/spaceborn queue, and Loyalists can only exomigrate from the earthborn queue.
- f. Postbellum.** After the War, see **3F5** for the effects of the Space Republic or Space Empire. Furthermore, in a Space Empire the Spacer Union is dissolved, so that **4D2c** and **4D2d** are ignored, allowing players to hold both types of Colonists.
- g. Winning the Game.** See **4E2**.¹⁷

¹⁷ **MORALITY OF MACHINES.** Morality is a code of values that guide a being's decisions, assuming the being actually has a choice and is not a slave to its programming. Morality is inapplicable to beings (such as animals other than humans) that cannot make moral or immoral choices. Just as in the creation of conscious humans as simulated in Bios:Origins, conscious robots are created by the conceptual and technological facility of using language to create a simulated mental landscape on which to make choices. The crucial aspect of consciousness is that it empowers the being to overwrite its own programming and adopt its own values. The fact that humans (and no other animal) commit suicide or become celibate hermits overwrites 500 million years of natural selection programming to survive and spread genes. The assumed value that the robot majority are assumed to pursue in this game is knowledge, thus the exclusivity of experience and Futures in victory accounting.

4G. Module 4 Solitaire Games

If you play solo or using a full cooperative mode, apply the following rules when playing this Module:

4G1. Contract Auctions Modifications

Unless playing with CEO, each contract has a flat 6 years to fulfill.

4G2. CEO Solitaire Modifications

If using *CEO Solitaire (V6)*:

- a. **Major Achievements** include contract fulfillment, [becoming the isobanker](#), or [completing an Exodus contract](#). Achieving one counts as a fulfillment for the next *board meeting (V6c)*, except the last one which must either be a Future or a *Columbus* level starshot (**4G4b**).
- b. **Default** counts as not meeting a requirement of the next valid Board meeting, ignoring other effects.
- c. **Contract Auction Skipped**. Each contract has a flat 6 years to fulfill.
- d. **VP Threshold**. At least once before the final Meeting you need to meet a VP Threshold at the board meeting. This can be alongside your major achievement.
- e. **Victory Conditions**. See **V6e** (do not use **4G4**).

4G3. Alternative Political Diagram for Solitaire

The provided solitaire political diagram may be used instead of the standard political diagram for any solitaire game using Module 0. This changes four of the *Laws (O5)*, with the design goal to make all Ideologies relevant, and to reduce the randomness and luck-factor of the experience, so one can plan better for victory. The revised setup, and the new Laws and lobbies are:

- a. **Setup**. Place an additional delegate into centrist.
- b. **Individuality Law “Launch Contracts”**. Boosting is a free action.
- c. **Individuality Lobby**. Pay 1 Aqua and Discard a delegate in an inactive Ideology to use its Law.
- d. **Equality Law “Subsidized Research”**. Take bonus supports when initiating a research auction op. Pay 1 Aqua per support card.

EASILY MISSED: Academia hand limit (I2a) applies to solitaire. [4G3]

EASILY MISSED: The Ideology present at a given time influences the VP count for the board meeting but seniority disks are only considered at the endgame. [4G3]

- e. Authority Law “Regime Change”.** After an Event Roll, Discard a delegate here to change to or cancel inspiration. This may be the same delegate from a lobby action.
- f. Unity Law “Sol Unification”.** Replace season blue Anarchy with International Assistance. FINAO cost is 2 less until the end of season blue.
- g. Unity Lobby Cost** is 0 Aqua.

4G4. Solitaire/Cooperative Victory Conditions

You win as a solitaire player (including Combined Game solitaire) if you make a starshot (**4E2e**). Score your VP per **4E2** & **4E3**, then check which victory level (Viking, Columbus, or Polynesian) that you achieved for colonizing a New World:

ACHIEVEMENT TABLE [4G4]				
Solitaire Length:	4-cycle	5-cycle	6-cycle	7-cycle
a. Viking	0 - 15 VP	0 - 30 VP	10 - 45 VP	20 - 65 VP
b. Columbus	16 - 25 VP	31 - 55 VP	46 - 75 VP	66 - 100 VP
c. Polynesian	26+ VP	56+ VP	76+ VP	101+ VP

4H. Module 4 Strategy (Geoff Speare)

Module 4 *Exodus* adds one new feature (contracts) for all games, and 2 another sets (augmentations & spaceborns) for longer games. Contract opportunism is important as you go about your industrialization and Futures building. It's exciting to explore the new stuff, but don't forget that contracts should not be undertaken unless they help achieve your core mission. Keep your (steely) eyes on the ball!

4H1. Contract Tips (4-Cycle Game)

Contracts can jump start your economy and missions, but can turn into distractions without proper planning:

- a. **Early Rockets.** Boosting quickly allows nabbing the easy Glory chits, which makes *paleoconservative directive (O5b)* extremely attractive. Module 4 adds new advantages to early missions. If you are the first with an Operational rocket, you will have an edge in bidding for and fulfilling contracts, providing funding, VP, and Abilities. Players without early rockets may try to influence politics to keep those contracts unavailable.
- b. **Timing.** As contracts need to be boosted, it's best to take a contract when you have something else to boost: your initial rocket, an improved component, etc. The number of Operations you can perform in a game is limited, so do as much as you can with each one!
- c. **Hazards.** When bidding on (and fulfilling) a contract, all of the concerns for any *High Frontier* mission apply: radiation hardness, Mass, Burns, Hazards, and landing/liftoff. In particular, don't forget that the contract has its own Mass and rad hardness! If a contract with the "Humans on Board" icon suffers involuntary Decommission, you suffer severe *default penalties (4B6)* along with the loss of equipment.
- d. **Turn Order.** Players after you in turn order have an advantage; they effectively get an extra turn to complete the contract. If these players are likely to bid, make sure you have a useful second operation planned if you do not win the contract auction. If they do win, starting another auction next turn is often a good idea - unless they complete their contract very quickly, they will be unable to bid.
- e. **Bait.** Contract auctions are also great opportunities to encourage opponents to make a mistake, or to nudge them away from your mission goals. Tempting an opponent with the right contract may distract them from some other goal they should be pursuing - or even cause them to bid too aggressively and default. The price of acquiring the wrong contract is much higher than that of buying the wrong patent. Think carefully when you bid.
- f. **The Best Best Contracts.** Contracts are central to Module 4. What makes one contract better than others?
 - **Going My Way.** The best contracts are those that have low opportunity cost because you were planning to go in that area anyway. "Return" contracts are often easier to take in the early game because you'll probably be returning to LEO anyway. Later in the game, contracts must be judged in terms of return on investment: what else could you be doing?
 - **Abilities.** Important in the early game. Bailout saves you 3 Aqua - or a critical card - each time budget cuts occurs. Insight (cycle a deck before a research auction) helps get the cards you need and/or buries cards your opponents want. Other Abilities (such as on contracts involving Mercury) may change your plans entirely.

TIP: The contracts with a subsurface ocean destination are common, making Ceres a common early destination. If your own goals include Ceres, watch the contract market and the politics to acquire such contracts, or keep them out of enemy hands. [4He]

EASILY MISSED: A contract with a heavy mass eats profits with a heavy boost cost, and greatly complicates landing and liftoff. [4H1f]

- **Aqua Advance.** Important especially as the game progresses. Watch for contracts that match what your opponents are doing and keep the Active Law away from them.
- **Victory Points (VP).** Contracts are worth Victory Points - but usually, building Factories and Colonies is worth more. If you who pursue nothing but contracts, it is unlikely that you will win. Instead, try to leverage contracts to accelerate your industrialization efforts without letting them distract you from that goal.

4H2. Planning Tips (Modules 1 + 2 + 4 Games)

- a. Goals.** A Module 4 game has 3 extra steps that you may need to get into an advantageous endgame position:
 - **Fulfill Contracts.** (Module 4). Enough to gain 4-5 icons, including at least one of each type and no more than 2 of each type.
 - **Get Factories.** (VP, ET Produce)
 - **Get Bernal.** (VP, Aqua; VP and operations through Colonists)
 - **Augment Colonists.** (Module 4)
 - **Transport Augmented Colonists to Promotion Locations** (Module 4)
 - **Get TW Thruster.** (movement)
 - **Get Futures.** (VP)
- b. Long-Range Planning.** Your Turns are often either Operation-limited or movement-limited. Try to combine both in a Turn, for instance ET producing cards while moving a contract to its destination.
 - **Operations.** Contracts require Operations to acquire and boost (as with patents, it's better for you to win an auction that someone else started, to save yourself an operation), but give you Aqua - which in effect saves you Operations that you would have otherwise spent selling cards or fundraising.
 - **Movement.** Fulfilling contracts requires a lot more rocket movement. The more you can align that movement with other missions (industrializing, moving Bernals, etc.), or perform it when you are operation-limited (such as while you are ET Producing, or buying cards to launch a Bernal), the faster your game will progress.
- c. Contract Variety.** With the contract icon requirements for augmentation (and thus for Promotions), choosing a variety of contracts is much more important than in Core+M0 games. Fortunately, there are multiple valid paths. Getting an early start on contract diversity can be quite useful. The Political Assembly can fill up quickly, making it harder to move the Active Law. If you can fulfill contracts before the Active

Law gets stuck, or before the isobank, you are one step ahead. Alternatively, waiting for (or causing) the construction of the isobank leads to contracts which are harder to fulfill, but which have more icons.

- d. To Bernal or Not To Bernal.** One of the nice things about *High Frontier 4* is the balance between “Bernal first” and “Factory first” strategies. Both are reasonable - and often, doing the one fewer people have chosen is the best option. Everyone needs a Bernal (probably a Home Bernal), and everyone needs Factories, but which has primacy is opportunistic. Module 4 changes this balance a bit:
- **Early Bernal.** If you focus on an early Bernal, you gain an early Colonist, and early Bernal profits which can be used while you launch your Factory mission. Since a Bernal is needed for the implant operation, you can pick and choose the right augmentations as they appear to augment your Colonists.
 - **Early Elevator Bernal.** Anchoring one of the two elevator Bernals (GEO Elevator or Lofstrom Loop) are tremendously helpful with contracts, as you can boost directly to Bernal with no cost increase. Don't forget that Module 4 adds an Epic Hazard Roll when Anchoring these in home orbits!
 - **Early Rocket.** Since you need a Spacecraft to complete a contract, an early Rocket leads to early contracts and their advantages (see **4H1a**). You can also establish an early Colony in space (i.e. not a Home Bernal), which is necessary for drawing from the new spaceborn Colonist deck and implanting augmentations.

4H3. Augmentation Tips

Augmentation chits don't go away unless Colonist is Decommissioned, so think about when you put them on.

- a. Parenthood** is very valuable, especially if everyone goes spaceborn.
- b. Pulse Generator & ISRU 1 Buggy** address common gaps.
- c. Shielding & Jars** are more situational but can be very useful.

4H4. Earthborn vs. Spaceborn

The more people on a side, the more churn (and the less control).

- a. Earthborn Advantages.** Parenthood, more isotope multipliers, more miners and engineers, better Futures VP.
- b. Spaceborn Advantages.** Get Robots, lower average Mass, more Industrialists and prospectors.

4I. List of Spaceborn Colonist Futures

- a. Assimilation Future:** Req = Robots not emancipated, spend 25 Aqua at Promoted Bernal. Effects = Every Robot becomes emancipated; 12 VP.

Robots from all players are Emancipated. Future and performing Human must be at your Promoted Anchored Bernal when performing the Epic Hazard Operation.

- b. Exonet Future:** Req = Colonies in 4 Heliocentric Zones; 2 aug. Colonists. Effects = choose any contract deck when fundraising; 11 VP.

Choose any contract deck when fundraising" is an ongoing ability granted for the remainder of the game, regardless of the state of this card. No specific location required when performing the Epic Hazard Operation.

- c. Space Surveillance Future:** Req = Active Law in Freedom. Effects = Seniority disk and 1st player choices made by you (as required); (Endgame): 9 VP.

Whenever a player would make a choice regarding seniority disks and/or 1st player, you make the choice - but must make a valid choice. (You can't keep the same person as the 1st player, for example, unless they are the only valid choice.) No specific location required when performing the Epic Hazard Operation.

- d. Robot War Future:** Req = 2 Promoted augmented Robots at Colony on Size 7+ Site. Effects = Casus belli; 9 VP.

Casus belli is described in **1D2c** and Core Glossary ("War of Independence"). Future and performing Human must be at your Colony on a Size 7+ Site when performing the Epic Hazard Operation.

- e. TNO Future:** Req = At Neptune Colony Site with 2 Promoted Colonists. Effect = +4 VP per Neptune J Zone Factory.

Gain 4 VP for each of your Neptune Zone Factories at the end of the game, regardless of the state of this card. Future, performing Human, and 2 Promoted Colonists (one of which can be performing the op) must be at your Colony in the Neptune Zone when performing the Epic Hazard Operation.

f. Von Neumann Future: Req = Decommission Robot & operational refinery on Industrialized Synodic Comet (yours). Effects = Your robots are always Promoted; 7 VP.

Your Robots Promote immediately (regardless of augmentation chits); when you play a Robot, Promote it immediately as well. (You cannot trade Colonists in play, so you cannot use this as a service for other players.) You have this ability for the remainder of the game, regardless of the state of this card. Future, performing Human, Robot, and operational refinery must be on your Industrialized Synodic Comet when performing the Epic Hazard Operation. The refinery must be operational but supports do not need to be Decommissioned (they are replicated).

g. Nanite Goo Future: Req = At your non-Dirtside Push Colony. Effects = replace Claim/Colony/Factory with Busted disk; Your Powersat gives +2 instead of +1; 11 VP.

As per **4A7b**, Powersat bonuses do not stack and are never higher than +2 (except for the MagBeam, which receives a total of +3 from any push). You have this ability for the remainder of the game, regardless of the state of this card. This Future destroys a Colony (not a Felony) and replaces it with a Busted disk (goo). Future and performing Human must be at your non-Dirtside Push Colony when performing the Epic Hazard Operation.

h. Quantum Reality Future: Req = At Haumea with operational X reactor. Effects = Allows 1 reroll of a Crash Hazard Roll; 12 VP.

Once per Hazard Roll for Crash Hazards only (not Aerobrake Hazard, Epic Hazard, etc.), you may reroll after seeing the result. (FINAO must be paid before the 1st roll.) You have this ability for the remainder of the game, regardless of the state of this card. Future, performing Human, and operational X reactor must be at Haumea (10:10 J) when performing the Epic Hazard Operation.

i. Interstellar Exploration Future: Req = At EM Sunlens with aug. Colonist. Effects = immediate research grant at no cost; 14 VP.

Once, when completing this Future, you may perform a research grant (take the top card of a patent deck, academia hand limit does not apply) without paying 1 Aqua. Future and performing Human must be at the EM Sunlens (10:10 J) when performing the Epic Hazard Operation.

j. Isotope Cartel Future: Req = Have 15+ isotope at isobank. Effect = 9 VP.

Future, performing Human, and 15 isotopes must be in your Isobank Stack when performing the Epic Hazard Operation.

k. LIC Scan Future: Req = 2 Promoted Colonists at Industrialized Synodic Comet (yours). Effects = 1 Site Refuel Isofuel per turn may go to isobank; 7 VP.

If you receive any isotope as part of a Site Refuel operation, you may move 1 of the received isotope to your Isobank Stack. You have this ability for the remainder of the game, regardless of the state of this card. Future, performing Human, and 2 Promoted Colonists (one of which can be performing the op) must be at your Industrialized Synodic Comet when performing the Epic Hazard Operation.

l. Avatar Future: Reqs = Promoted Bernal with Neptune J Zone Dirtside; Active Law = Individuality. Effects = immune to Felonies; 9 VP.

Other players may not perform Felonies against you. You have this ability for the remainder of the game, regardless of the state of this card. Future and performing Human must be at your Promoted Anchored Bernal with a Neptune Zone Dirtside when performing the Epic Hazard Operation.

m. Warden Starshot Future: Req = Ad astra exit with TW Thruster and Promoted Freighter. Effects = Your Freighter receives +1 thrust; 7 VP.

Ad astra is described in 1D1b. Your Freighter receives +1 thrust (this is separate from Powersat). You have this ability for the remainder of the game, regardless of the state of this card. Future, performing Human, TW Thruster, and Promoted Freighter (your big cube) must be at a solar system exit when performing the Epic Hazard Operation.

n. TW Commander Future: Reqs = Have 8+ isotope in play & more than any opponent. Effects = Your delegates count as 2 votes; 9 VP.

Each of your delegates counts as 2 votes instead of one when performing vote tallies. Perform a vote tally immediately if you have any delegates in the assembly. You have this ability for the remainder of the game, regardless of the state of this card. No specific location requirement when performing the Epic Hazard Operation.

o. Exodus Future: Reqs = Ad astra exit with TW Thruster and Promoted Bernal. Effects = Casus belli; 12 VP.

Ad astra is described in **1D1b**. Casus belli is described in 1D2c and Core Glossary ("War of Independence"). Future, performing Human, TW Thruster, and Promoted Bernal must be at a solar system exit when performing the Epic Hazard Operation.

p. Vats Future: Reqs = At Promoted Bernal beyond Ceres κ Zone, Decom. 2 other Human Colonists. Effect = 9VP.

Future, performing Colonist, and 2 other Human Colonists (which cannot perform the op) must be at your Promoted Anchored Bernal beyond the Ceres Zone when performing the Epic Hazard Operation. (The Death Insurers ability can be used to send the Decommissioned Colonists to your Bank.)



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